

# Airfoil Design Framework for Optimized Boundary-Layer Integral Parameters

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An airfoil design framework is introduced in which boundary-layer integral parameters serve as the driving design mechanism. The method consists of generating a parameterized pressure distribution capable of producing the desired boundary-layer characteristics for inverse design use. Additionally, by deduction from the Squire–Young theory, the method allows for the determination of the pressure distribution that results in the minimum theoretical drag. To assess this design framework, several airfoils were developed based on the mission requirements of the RQ-4B Global Hawk aircraft. Numerical results obtained using a viscous-inviscid solver of the integral boundary layer and Euler equations showed that the optimized airfoils achieved profile drag reductions of 9.06 and 6.00%, respectively, for  $\alpha = 0^\circ$  and  $L/D_{\max}$  design points. A validation experimental campaign was also performed using the optimized CA5427-72 airfoil. The acquired data produced the expected pressure distribution characteristics and aerodynamic performance improvements, typifying the efficacy of the design framework.

## Nomenclature

|                   |   |                                                      |
|-------------------|---|------------------------------------------------------|
| $C_D$             | = | dissipation coefficient                              |
| $C_d$             | = | profile drag coefficient                             |
| $C_{f,x}$         | = | skin friction coefficient                            |
| $C_l$             | = | lift coefficient                                     |
| $C_{l_{\max}}$    | = | maximum lift coefficient                             |
| $C_m$             | = | pitching moment coefficient about aerodynamic center |
| $C_p$             | = | pressure coefficient                                 |
| $C_{p_{TE}}$      | = | pressure coefficient at trailing edge                |
| $C_\tau$          | = | shear stress coefficient                             |
| $C_{\tau_{EQ}}$   | = | equivalent shear stress coefficient                  |
| $c$               | = | airfoil chord                                        |
| $H$               | = | shape factor                                         |
| $H_k$             | = | kinematic shape parameter                            |
| $H^*$             | = | kinetic energy shape parameter                       |
| $H^{**}$          | = | density shape parameter                              |
| $L/D$             | = | lift-to-drag ratio                                   |
| $L/D_{\max}$      | = | maximum lift-to-drag ratio                           |
| $l, m$            | = | empirical formulas                                   |
| $M$               | = | Mach number                                          |
| $M_{\text{crit}}$ | = | critical Mach number                                 |
| $M_e$             | = | boundary-layer edge Mach number                      |
| $M_\infty$        | = | freestream Mach number                               |
| $n$               | = | amplification factor                                 |
| $n_{\text{crit}}$ | = | critical amplification factor                        |
| $Re$              | = | Reynolds number                                      |
| $Re_\theta$       | = | momentum thickness Reynolds number                   |
| $t/c_{\max}$      | = | maximum thickness-to-chord ratio                     |
| $u_e$             | = | boundary-layer edge velocity                         |
| $u_\infty$        | = | freestream velocity                                  |
| $x$               | = | streamwise coordinate                                |
| $x_0$             | = | Stratford and pressure recovery onset location       |

|                 |   |                                                        |
|-----------------|---|--------------------------------------------------------|
| $x_1$           | = | minimum pressure location                              |
| $x_2$           | = | transition location                                    |
| $\alpha$        | = | angle of attack                                        |
| $\delta$        | = | boundary-layer thickness                               |
| $\delta^*$      | = | boundary-layer displacement thickness                  |
| $\delta_{TE}^*$ | = | boundary-layer displacement thickness at trailing edge |
| $\theta$        | = | boundary-layer momentum thickness                      |
| $\theta_{TE}$   | = | boundary-layer momentum thickness at trailing edge     |
| $\xi$           | = | shear layer coordinates                                |

## I. Introduction

**A**IRFOIL design has been of interest since the early days of aviation, when flight pioneers sought inspiration from birds, recognizing the importance of their wing sections for sustaining flight. Since then, the field has matured in such a way that standard methods are in place for the design and analysis of airfoil sections used on aerodynamic vehicles and other applications. Airfoil sections are simply characterized by their lift, drag, and pitching moment performance, which is highly dependent on physical design properties (e.g., thickness, pressure distribution, and surface orientation) as well as on their operating conditions (e.g., Mach and Reynolds numbers). For this reason, new airfoil design methods and approaches are still of great interest, as they can be utilized to enhance the aerodynamic efficiency for specific applications and operating regimes.

Inverse design is a common method used for the generation of airfoil sections. Through this approach, the designer specifies general aerodynamic characteristics a priori to obtain the corresponding airfoil geometry that satisfies such constraints. One of the first design methods that showed great success was that developed by Lighthill [1] using conformal mapping, which only required a velocity distribution to be defined. Lighthill went on to suggest that the benefit of such a design tool would be to control boundary-layer properties by imposing conditions on the velocity distributions through classical potential theory. Perhaps the most canonical example of conformal mapping used for the generation of airfoils is the Joukowski transform. In this approach, Joukowski began with the potential flow solution past a cylinder and conformally mapped the circular shape onto a flat plate using the complex plane, resulting in an airfoil geometry [2]. Other single-point inverse design methods still function under the same basic principle of mathematically generating a geometry based on potential pressure or velocity constraints.

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Another common approach used in airfoil design is the multipoint methodology, which aims to satisfy multiple operating requirements at different angles of attack. This theory, which was first presented by Eppler [3,4] and later expanded by Selig [5,6], requires the designer to define a velocity-function or velocity-constant segment along the airfoil surface at specific angles of attack. Defining the velocity constraints in this manner allows the multipoint design objectives to be satisfied simultaneously throughout the airfoil generation process. While the multipoint theory offers a more direct and inexpensive airfoil design approach, single-point inverse design methods such as those initially introduced have still been successfully used to generate airfoil geometries that also achieve off-design operational constraints [5]. Furthermore, the single-point inverse design process provides better direct control of the boundary-layer behavior through the specification of a desired velocity or pressure distribution [7]. The greater measure of boundary-layer control offered by this method is also likely to have the greatest capacity to further enhance the aerodynamic performance of air vehicles at their most critical operating condition. It should be noted, however, that performance gains at a single design point may come at the expense of performance at other conditions.

A very useful and effective inverse design methodology that has been used for airfoil design is that introduced by Giles and Drela [8]. This method consists of the simultaneous solution of multiple streamtubes that are coupled to the location and pressure values of the streamline interfaces. This process allows the airfoil shape to be specified in a direct design mode and, perhaps of most interest, the pressure distribution in the inverse design mode. In addition, the method is formulated in a conservative manner, such that airfoil flows with shocks in the transonic environment are simulated effectively. More specifically, this inverse design tool is coupled with the viscous-coupled solver of Drela and Giles [9] into the well-known program MSES, which allows the geometry that produces a desired pressure distribution to be simultaneously identified through the flow solution using a global Newton method. The mixed-inverse design routine also gives the option to include viscous effects through the coupled integral boundary-layer method featured in MSES. To validate this methodology, a transonic airfoil was designed for operation at  $M = 0.70$  and  $C_l = 1.0$  using a seed geometry, and its performance was improved at off-design conditions through the specification of a pressure distribution that significantly reduced the shock intensity [8].

Integral boundary-layer methods have traditionally been coupled to inviscid solutions as a means to provide predictions of the viscous aerodynamic characteristics of airfoils, as previously described in the case of MSES, for example. However, different integral boundary-layer methods have also been successfully used in a more general airfoil design framework. For instance, the geometry that is obtained from an inverse design method is expected to produce the velocity or pressure distribution that was originally prescribed. Such a condition can typically be achieved for an inviscid solution where the velocity is governed directly by the airfoil contour. However, some definitions of pressure distributions can lead to airfoil sections that even though mathematically correct, are not realistic since they feature intersecting curves, for example. In part, this problem results from blindly defining pressure distributions without considering the viscosity effects within the boundary layer close to the airfoil surface that alter the apparent geometry as would be observed by a potential flow. To address this challenge, Henderson [10] considered the incorporation of an inverse boundary-layer technique to the inverse design process of airfoils. This approach consisted of defining a desired shape factor and its type of variation along the pressure recovery region to obtain its corresponding incompressible pressure distribution [10]. Even though the boundary layer, and hence the performance of the airfoil, could be defined to a limited extent, the objective of this study was not primarily to develop novel airfoil designs for desired performance. Rather, the primary purpose was to obtain pressure distributions that ensured realistic sections would be obtained from the inverse design process by considering boundary-layer parameters.

It is well known that the performance of an airfoil is highly dependent on the boundary-layer characteristics produced at a certain operating condition. Therefore, the primary purpose of this study is to introduce a new airfoil design framework that considers the boundary-layer parameters, namely  $\delta^*$ ,  $\theta$ ,  $C_{fx}$ , and  $n$ , to drive the design of an airfoil. The method being introduced consists of the generation of a parameterized pressure distribution that is capable of producing desired boundary-layer characteristics, which is then used to obtain a corresponding airfoil geometry through an inverse design routine. First, the design framework is introduced alongside underlying assumptions and implications. Next, a family of airfoils that were developed using the proposed design methodology is presented with the intent of highlighting the advantages associated with a design framework that incorporates integral boundary-layer parameters. The mission profile of a high-altitude, long-endurance (HALE) unmanned vehicle was considered to define target airfoil design constraints. The MSES theoretical results of the airfoil geometries, for which different target operating conditions and boundary-layer design constraints were considered, are then discussed in great detail. Lastly, the experimental results are analyzed for the CA5427-72 airfoil, which was specifically designed for HALE operating constraints that could be achieved in an experimental environment to validate the design method being proposed.

## II. Airfoil Design Methodology

### A. Overview

The design methodology for the current study considers the boundary-layer performance as the driving mechanism for the airfoil inverse design process. In particular, the pressure distribution is configured to achieve a desired parameter condition at the trailing edge (e.g.,  $\theta_{TE}$  and  $\delta_{TE}^*$ ) that ultimately dictates the upstream skin-friction distribution and influences the airfoil drag characteristics and off-design performance. For instance, the design of laminar flow airfoils commonly features extensive regions of favorable pressure gradient (FPG) across most of the upper surface to delay boundary-layer transition and minimize the skin friction drag component. The pressure recovery of these airfoils across the adverse pressure gradient (APG) region is usually designed to occur over a short distance and also have a low skin-friction characteristic, such as that produced by the Stratford distribution [11–14]. Nevertheless, a factor that might be overlooked in trying to minimize the skin friction drag component, particularly across the recovery region, is the growth of the boundary-layer displacement thickness, which could lead to higher pressure drag. Hence, a method that considers both skin friction and pressure drag contributions from any given pressure distribution based on the trailing edge boundary-layer parameters would aid in determining the distribution that results in the minimum profile drag condition.

A key component in the methodology being considered is the tailoring of the boundary-layer development through the variation of a parameterized pressure distribution. Thus, an integral boundary-layer code is incorporated into the design routine to help identify the optimal inviscid pressure distributions based on the predicted boundary-layer states. These types of methods have been traditionally used to predict the aerodynamic characteristics of airfoils when coupled to inviscid flow solutions. In lieu of carrying out an exact solution of the boundary-layer differential equations for every fluid particle, integral methods present an approximation of the mean boundary layer by satisfying the boundary conditions at a single location near the wall [15]. This can be performed since the Prandtl boundary-layer formulation shows that the pressure gradient normal to the surface is negligible, and therefore the airfoil pressure gradient resulting from an inviscid solution can be considered to be a driving parameter in the boundary-layer development [16]. Consequently, the approximation is very useful when the pressure distribution can be predicted separately through means of an inviscid analysis method [16]. Examples of such analysis tools include PROFIL, PROFOIL, XFOIL, and MSES [9,17–19], all of which are coupled to an integral boundary-layer solver.

Another canonical application of the integral boundary-layer theory for the direct calculation of profile drag is that presented by Squire and Young [20]. By considering a control volume analysis of the wake far downstream of the airfoil where the static pressure equaled freestream conditions, Squire and Young related the profile drag calculation to the momentum thickness of the wake at the same freestream location. The momentum thickness of the wake at the airfoil trailing edge was then assumed to be simply the sum of the momentum thickness of the upper- and lower-surface boundary layers. Through the use of the momentum equation, the far downstream wake momentum thickness was then calculated from the momentum thickness at the airfoil trailing edge and used to predict the profile drag [20]. From this theory it can be deduced that to obtain the minimum profile drag generated by an airfoil for a fixed  $C_l$  condition, the value of  $\theta$  at the trailing edge of both surfaces must be minimum. Hence, identifying the pressure distribution that minimized  $\theta_{TE}$  through an integral boundary-layer solution became the primary objective of this study.

As is common to airfoil design problems, the first step in the current methodology is to identify the relevant flow operating conditions and target airfoil performance parameters such as Mach number, Reynolds number, and lift coefficient. The upper surface of the airfoil is then assumed to consist of three distinct segments, including an initial FPG, a region of linear pressure variation, and subsequent pressure recovery. An array of pressure recovery onset locations are then considered based on a  $C_l$  constraint to generate different pressure distributions with varying transition locations. The family of pressure distributions generated is then analyzed using the integral boundary-layer code developed, considering both the laminar and turbulent regions. The boundary-layer solutions are then analyzed to identify the pressure distribution that provides the most preferred boundary-layer condition at the trailing edge of the airfoil. Once the target  $C_p$  distribution is selected, the airfoil geometry that produces such a distribution is obtained through an inverse design routine. Figure 1 shows a flow diagram outlining the different steps and processes in the design methodology just mentioned, which are explained in greater detail in the following subsections.

## B. Integral Boundary-Layer Method

The boundary-layer characteristics of the airfoil pressure distributions were modeled via a two-equation integral method based on the formulation provided by Drela and Giles [9]. The equations for the integral momentum and kinetic energy shape parameters are as follows [9]:

$$\frac{d\theta}{d\xi} + (1 + H - M_e^2) \frac{\theta}{u_e} \frac{du_e}{d\xi} = \frac{C_{f,x}}{2} \quad (1)$$

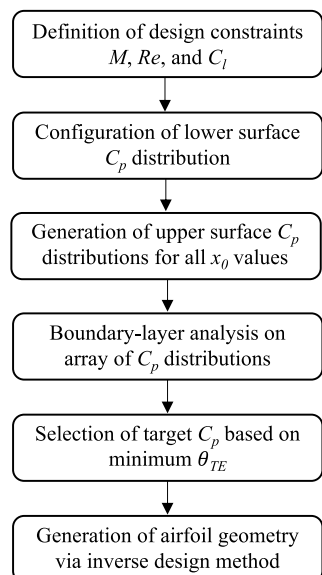


Fig. 1 Flow diagram for the airfoil design process being proposed.

$$\theta \frac{dH^*}{d\xi} + [2H^{**} + H^*(1 - H)] \frac{\theta}{u_e} \frac{du_e}{d\xi} = 2C_D - H^* \frac{C_{f,x}}{2} \quad (2)$$

This set of equations assumes a dissipation type closure in which terms  $H^*$ ,  $H^{**}$ ,  $C_{f,x}$ , and  $C_D$  are shown to have dependencies on  $H_k$ ,  $M_e$ , and  $Re_\theta$ . For the laminar boundary-layer closure, a Falkner–Skan self-similar profile is employed to derive the terms in Eqs. (1) and (2). These terms are obtained for the turbulent boundary-layer closure by considering the skin-friction and velocity profile formulas of Swafford [21]. The expressions for the closure parameters used can be found in Refs. [22,23].

In comparison to the laminar boundary layer, the dissipation coefficient for the turbulent boundary layer has an additional dependency on the shear stress coefficient  $C_\tau$ . Furthermore,  $C_\tau$  depends on transport behaviors associated with the Reynolds stresses in the wake layer, which respond relatively slowly to changing conditions. This behavior is modeled through the following rate equation [19], which has been modified from that originally presented in Ref. [9] and found to produce better predictions near stall:

$$\frac{\delta}{C_\tau} \frac{dC_\tau}{d\xi} = 5.6 \left( C_{\tau_{eq}}^{1/2} - C_\tau^{1/2} \right) + 2\delta \left\{ \frac{4}{3\delta^*} \left[ \frac{C_{f,x}}{2} - \left( \frac{H_k - 1}{6.7H_k} \right)^2 \right] - \frac{1}{u_e} \frac{du_e}{d\xi} \right\} \quad (3)$$

Hence, the solution for the turbulent boundary layer depends on the three-equation coupled system consisting of Eqs. (1–3).

The transition location was also considered to properly model the boundary-layer development across the airfoil pressure distributions. Prediction of this location was performed by applying spatial-amplification theory to the Orr–Sommerfeld equation, which results in the ubiquitous  $e^n$  method. This method assumes that the boundary layer transitions once the most unstable Tollmien–Schlichting (T-S) wave in the boundary layer has grown by a factor of  $e^n$ , where  $n$  is an empirical constant [9]. Traditionally, a value of  $n = 9$  is used as it is representative of the low-freestream turbulence intensity encountered in atmospheric flight. The solution for the amplification factor  $n$  is obtained through the following differential equation [9]:

$$\frac{dn}{dx}(H_k, \theta) = \frac{dn}{dRe_\theta}(H_k) \frac{m(H_k) + 1}{2} l(H_k) \frac{1}{\theta} \quad (4)$$

which depends of the following empirical formulas [9]:

$$\frac{dn}{dRe_\theta}(H_k) = 0.01 \{ [2.4H_k - 3.7 + 2.5 \tanh(1.5H_k - 4.65)]^2 + 0.25 \}^{1/2} \quad (5)$$

$$l(H_k) = \frac{6.54H_k - 14.07}{H_k^2} \quad (6)$$

$$m(H_k) = \left( 0.058 \frac{(H_k - 4)^2}{H_k - 1} - 0.068 \right) \frac{1}{l(H_k)} \quad (7)$$

Similar to the turbulent boundary layer, the laminar boundary-layer solution depends on the three-equation coupled system comprised of Eqs. (1), (2), and (4).

The actual solution of the nonlinear, ordinary differential system of equations is carried out using a Newton numerical integration method through a code developed in MATLAB. The equations are discretized using a Crank–Nicolson numerical integration scheme in the form provided by Smith et al. [24]. This finite-difference scheme is second-order, implicit, and numerically stable, which provides better consistency for varying step sizes. The method simply requires that an edge-velocity ( $u_e$ ) distribution is provided (obtained from the  $C_p$  distribution), as well as initial conditions for  $\theta$ ,  $\delta^*$ , and  $C_\tau$ . These values are obtained from a viscous MSES solution of the initial seed airfoil, which is typically a fair assumption considering that these

parameters are initially close to zero and both the seed and modified airfoils are expected to have very similar pressure gradients across their leading edge regions. Lastly, the code solves for the distributions of  $\theta$ ,  $\delta^*$ , and  $C_\tau$  across the chord, from which all other important boundary-layer parameters can be determined.

### C. Pressure Distribution Parameterization

As is common in the operation and design of airfoils, the upper surface was assumed to serve the predominant role in dictating the net airfoil loads. Therefore, the optimization of the upper surface pressure distribution was only considered in this study, and the lower surface pressure distribution was designed to remain fixed. However, the current design framework could also be utilized across this surface to further extend upon results presented here. The upper surface distribution was assumed to consist of three main segments: an initial FPG, a linear distribution, and a concave pressure recovery region, which are shown in Fig. 2. Specific details regarding the design of each segment are provided in the following subsections.

#### 1. Segment I: Favorable Pressure Gradient

The first segment in the pressure distribution considers an FPG up to a specified chordwise location ( $x_1$ ) as observed in Fig. 2. The distribution across this region was designed to provide a minimum  $\theta$  growth that incurred the smallest possible profile drag contribution at the beginning of segment II. More specifically, the distribution was obtained by implementing an optimization routine on the boundary-layer solver discussed in Sec. II.B, where the laminar boundary-layer formulation was considered to be the objective function. The non-linear optimization function in MATLAB, *fmincon*, was used to obtain the pressure distribution up to the point where the local gradient of the distribution approached a value of zero. This function employs a gradient-based method and is designed to identify the minimum of a problem for which both objective and constraint functions are continuous and have continuous first derivatives. Alternatively, when the problem is unfeasible, the *fmincon* function identifies a solution that minimizes the maximum constraint value [25]. A constraint that the local slope (between two adjacent points in the Newton-system domain) had to be equal to or less than the previous was implemented in the optimization to ensure that the resulting pressure distribution would provide a feasible airfoil geometry and a  $\theta$  distribution with a positive growth rate. Figure 2 shows a representation of the optimized FPG region at a given operating condition. The location of  $x_1$ , which is associated with a pressure value in the optimized FPG distribution, is not directly selected but rather chosen to meet the design  $C_l$  constraint based on the distributions in segments II and III.

#### 2. Segment II: Linear Distribution

The second segment considers a linear pressure distribution up to the onset of pressure recovery. Furthermore, the pressure gradient across this segment ensures that transition occurs at a desired location using the  $e''$  method to prevent boundary-layer separation. Figure 2 shows the transition location  $x_2$  occurring upstream of the pressure recovery onset location  $x_0$ . The transition can also be designed to coincide with the pressure recovery onset location; however, this is likely to result in very aggressive recoveries close to separation limits that present convergence difficulties. In addition, placing the transition location slightly upstream prevents the risk of boundary-layer separation when the interface between segments II and III is blended to avoid discontinuities in the airfoil generation. The gradient across this linear segment is defined from  $x_1$  to  $x_0$  and its value is determined through an iterative process using the boundary-layer solver discussed in Sec. II.B, such that a transition is predicted to occur at the  $x_2$  location.

Airfoils designed to produce extensive regions of laminar flow generally feature regions of FPG that are known to inhibit the growth of T-S waves that instigate the natural boundary-layer transition process. Counterintuitively, Fig. 2 displays a distribution with an APG that leads to a natural transition at a chordwise location similar to those achieved by laminar flow airfoils designed with regions of FPG. This is due to the stabilizing influence effects imposed by compressible Mach numbers on T-S disturbances due to greater acceleration and change in density observed in the boundary layer [26–28]. A lower freestream Mach number would therefore require a higher FPG, resulting in a different distribution than that observed in Fig. 2. In addition, the chordwise Reynolds number has a role in dictating the natural transition location. Hence, for a particular operating Mach number, a higher chordwise Reynolds number condition will be associated with an increasingly high FPG across segment II in order to delay transition up to the desired location. Nevertheless, the current framework can still be used for these higher  $Re$  applications as well, which, similar to a low-Mach-number design, would result in a higher FPG than that shown in Fig. 2.

#### 3. Segment III: Pressure Recovery

The third segment consists of the pressure recovery region, which is critical in dictating the attached condition of the boundary layer and, as a result, ensures that the static pressure returns to the approximate freestream value. Initially, the Stratford pressure recovery distribution was considered as it provides the minimum skin friction contribution to the profile drag by producing a flow in which the boundary layer is on the verge of separation, where the  $C_{f,x}$  distribution asymptotes to zero [13,14]. First used on airfoil design by Liebeck, the Stratford recovery was observed to provide the shortest unassisted chordwise recovery region, which allowed a pressure distribution to be configured that achieved the theoretical maximum lift for a single-element airfoil. From experimental observations that proved the ability of this pressure distribution to produce recoveries with minimal skin friction, Stratford and Liebeck went on to suggest that its application was also likely to result in airfoils that produced minimum drag [12,14]. However, a practical aspect that also requires consideration is that the pressure distribution required to produce the marginally separated boundary layer across a short chordwise distance results in a substantial growth of the boundary layer when compared to other types. The rapid momentum loss across the boundary layer augments the growth of the displacement thickness, which alters the effective camber observed by the potential flow, limiting the pressure recovery magnitude. Hence, the Stratford distribution reduces the skin friction drag to a minimum at the cost of increasing the pressure drag, which could potentially offset any net profile drag benefits. In order to identify what type of pressure recovery provided the minimum growth in  $\theta$ , a parametric study was performed in which the degree of concavity was varied from a Stratford distribution to a linear distribution. Only concave pressure recoveries were considered, as they better prevent separation by decelerating the flow more aggressively when the boundary layer has more energy [27]. In addition, the Stratford distributions were

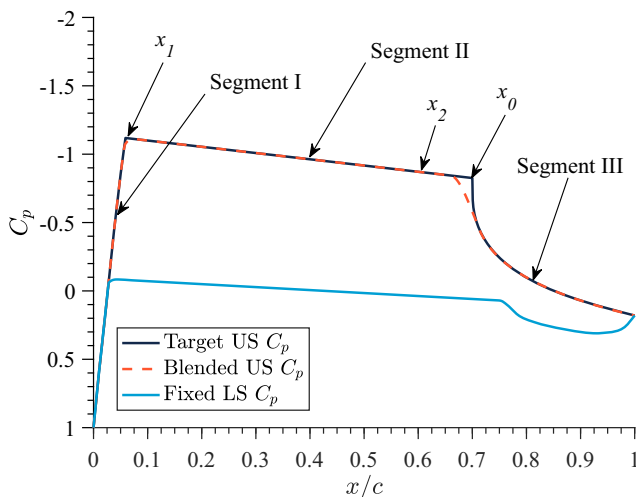


Fig. 2 Pressure distribution segments considered in the airfoil design methodology.

modified by stretching them in such a way that the onset of recovery would begin upstream of the Stratford theoretical limit. The variations in the pressure recoveries considered are presented in Fig. 3a.

The edge-velocity distributions, used by the boundary-layer solver discussed in Sec. II.B, were calculated for the different recoveries and used to predict the corresponding distributions of  $\theta$ . As observed in Fig. 3, a selected example of a modified Stratford distribution stretched to recover  $0.20c$  upstream of the Stratford limit produced the lowest growth in  $\theta$  at the trailing edge when compared to the other less concave recoveries. This behavior results from the modified Stratford having the highest growth in  $\theta$  near the recovery onset, which then asymptotes to a linear distribution when compared to the less concave recoveries in which the  $\theta$  growth displays a parabolic-type behavior across the recovery region. The better  $\theta_{TE}$  performance identified for the modified Stratford in contrast to the other recovery types was consistent regardless of the stretch factor applied. Furthermore, Fig. 4 shows that a greater stretch factor for a Stratford distribution will also result in a lower  $\theta_{TE}$  condition. However, it is evident that the pressure distributions in Fig. 4a would not result in the same  $C_l$  condition if a constant lower surface distribution for an airfoil were assumed. This observation suggests that the stretch factor applied to a modified Stratford recovery on an airfoil will be determined based on the lift and transition location constraints. Additionally, it can be expected that these Stratford-type recoveries will limit the maximum operating angle of attack of an airfoil as more aggressive pressure gradients upstream of the recovery produce thicker boundary layers

that are unable to withstand further inertial opposition and result in separation.

#### 4. Pressure Distribution Generation Algorithm

The generation of the pressure distributions consisting of the three segments previously described is an iterative process that considers several constraints. First, a gradient for the linear region of segment II and a corresponding  $C_p$  value at  $x_1$  are assumed. With these parameters,  $x_1$  and the  $C_p$  at the  $x_0$  location being considered are calculated. A Stratford distribution is then computed from the  $C_p$  at  $x_0$  to a defined  $C_{p_{TE}}$  using the code and solution process outlined in Ref. [29]. The resulting Stratford pressure distribution is stretched by the necessary factor for the recovery to extend from  $x_0$  to the trailing edge of the airfoil. All three segments are then combined to produce the target upper surface distribution shown in Fig. 2. The distribution is then interpolated at the segment interfaces, also shown in Fig. 2, as the blended distribution. This step was implemented to ensure a smooth transition between the different segments and avoid any discontinuities that would prevent the generation of realistic airfoil geometries. The interpolation was performed using a cubic Hermite spline algorithm since this method ensures a continuous function and first derivative without altering the intended form of the pressure distribution [30].

Also shown in Fig. 2, the prescribed lower surface distribution is then considered to calculate the  $C_l$  value that would be produced by the pressure distribution. If the calculated  $C_l$  is not within a defined

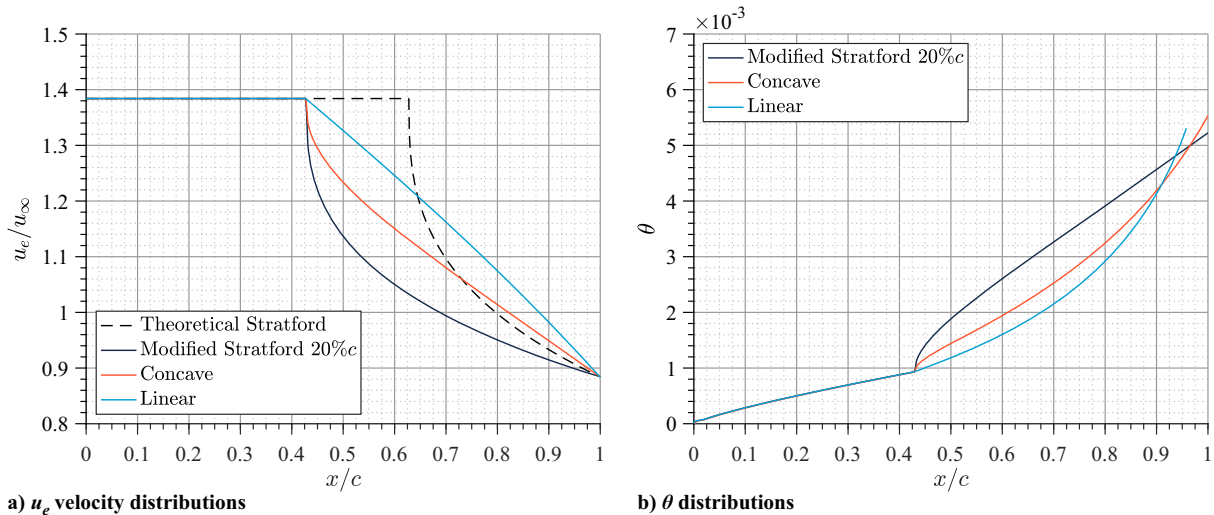


Fig. 3 Example of pressure recovery distributions considered where  $x_0$  is stretched 20%c upstream of the Stratford limit.

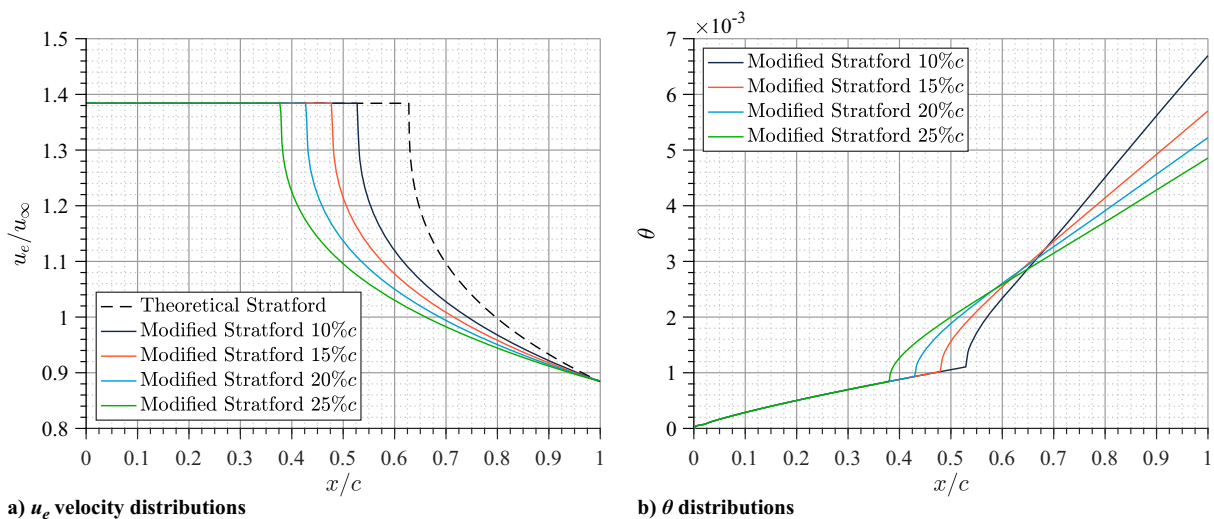


Fig. 4 Modified Stratford pressure recovery distributions with different chordwise stretch factors.



threshold of the target condition, a higher or lower value of  $C_p$  at  $x_1$  is determined and used to generate a new target distribution accordingly. This intermediate step shifts the distribution of segment II to higher or lower pressures and determines the corresponding distributions in segments I and III until the  $C_l$  threshold is met for the  $x_0$  constraint being enforced.

At this stage, the pressure distributions across segments I and II are analyzed using the boundary-layer solver developed for this study and discussed in Sec. II.B to determine the transition location. If the predicted transition location is not within a defined threshold of  $x_2$ , or the boundary layer fails to transition, the gradient across segment II is updated accordingly. The process of determining the target distribution considering the  $C_l$  and  $x_0$  constraints is repeated until the additional transition constraint at  $x_2$  is met. Target pressure distributions can be generated based on any  $x_0$  location that is upstream of the theoretical limit of a Stratford distribution, after which a shorter recovery would result in separation.

### D. Inverse Design Process

Target  $C_p$  distributions based on desired boundary-layer characteristics were used to obtain airfoil geometries using the Euler viscous-coupled solver MSES [9]. This software, also capable of effectively simulating airfoil flows within the transonic environment, features a mixed-inverse design routine that allows the geometry that produces a desired pressure distribution to be simultaneously identified with the flow solution [8,31].

The inverse design was performed using an inviscid analysis at the specified  $M_\infty$ , similar to that presented by Olson [31]. The process began by obtaining a viscous solution (at the operating  $Re$  condition and with forced transition at  $x_2$ ) of a seed airfoil with comparable performance characteristics to the one desired. The target  $C_p$  distribution previously designed was then used for the inviscid mixed-inverse design routine. Initially, the upper and lower surfaces downstream of the pressure recovery onset location were modified independently. The regions upstream of the pressure recovery up to the leading-edge suction peaks were then considered, also independently for the upper and lower sections. After each modification, a new viscous solution of the updated geometry was obtained before performing further inverse design iterations. This step ensured that the current flow solution being used to modify the geometry was representative of the viscous decambering effect observed across an airfoil surface. The process was repeated until the viscous solution of the modified airfoil matched the target  $C_p$  distribution. It is worth mentioning that, as is common for inverse design methods, geometry modifications close to the leading edge due to significant target pressure differences are usually unattainable or can lead to unrealistic geometries. Therefore, the design must begin with a leading edge that already matches the desired behavior near that region. Once a converged design was obtained, the coordinates were smoothed using a fourth-order B-spline to ensure slope continuity along the airfoil curve.

## III. Airfoil Development

To evaluate the airfoil design methodology being proposed, a HALE aircraft was considered to identify an exemplary set of airfoil design constraints and operating conditions. HALE aircraft, designed to operate for long periods of time at target altitudes of up to 35 km [22], have gained research interest in aerospace design sectors as they offer less expensive and serviceable solutions for intelligence, surveying, and reconnaissance (ISR), communications relaying, weather and agricultural monitoring, and other applications requiring satellite-based structures [32–37]. These aircraft are characterized by low wing loadings, which pose a need for highly efficient airfoil sections at an unconventional condition of low Reynolds and compressible-subsonic Mach numbers. Numerous design efforts have been tailored to HALE vehicles; however, the airfoil sections typically result in high  $t/c_{\max}$  from the low-density operating constraint rendering  $M_{\text{crit}}$  to low values. In addition, the performance of these airfoils is often limited by the presence of laminar separation bubbles (LSBs) [22,32]. Hence, the method being considered in this

study can be used to improve the performance of airfoil sections that meet the stringent operating conditions encountered by HALE vehicles.

Table 1 shows the general operating conditions of the Northrop Grumman RQ-4B Global Hawk, which was selected to define the airfoil performance requirements based on its mission profile. The LRN1015 airfoil used on the RQ-4B Global Hawk was also used as the seed geometry from which the optimized airfoils were generated. To further showcase the applicability of the design framework across a wider operating range, two additional design cases were considered, consisting of high  $Re$  and incompressible  $M$  conditions. A summary of all the airfoils developed in this study is presented in Sec. III.A. Section III.B outlines an example of the airfoil design process using the CA5427-72 airfoil, which was specifically designed for experimental validation. It should also be emphasized that the airfoils developed in this study have a point design constraint that limits their off-design performance when compared to the baseline seed geometry. Nevertheless, the intent of these designs is to showcase the implementation of and highlight the advantages associated with an airfoil design methodology that leverages integral boundary-layer parameters as the main design driver. Lastly, additional discussion on the development of the remaining airfoils, as well as geometrical coordinates, can be found in Ref. [38].

### A. Summary of All Airfoils

Table 2 shows a summary of all the airfoils developed in this study, alongside their corresponding design constraints. Target  $C_l$  parameters for all airfoils were obtained by performing an MSES solution on the LRN1015 airfoil at the desired operating condition (e.g.,  $\alpha = 0^\circ$  and  $L/D_{\max}$ ). For the CA5488 airfoils,  $\alpha = 0^\circ$  was selected as it would be representative of a normal-cruise flight condition. The target  $M$  and  $Re$  conditions for the CA5488 airfoils were chosen using the performance parameters in Table 1 and assuming a mean geometric chord. To assess the effect of optimizing target pressure distributions using different integral boundary-layer parameters, the CA5488-88 was configured for minimum  $\theta_{\text{TE}}$  and the CA5488-76 was configured for minimum  $\delta_{\text{TE}}^*$ . The lower surface for these airfoils was designed to resemble that of the LRN1015, with a natural transition predicted at the  $0.70c$  location. As previously mentioned in Sec. II.C, this lower surface pressure distribution remained fixed throughout the design optimization. Lastly, as can be observed in Table 2, the maximum freestream Mach number considered was 0.54, which, while compressible, produced shockless upper surface distributions. Based on previous work that explored the behavior of Stratford pressure distributions in compressible transonic flows, it is expected that this methodology can be effectively used at higher

**Table 1 Operating parameters for the RQ-4B Global Hawk [39]**

| Parameter       | Value               |
|-----------------|---------------------|
| Service ceiling | 60,000 ft           |
| Loitering speed | 310 kt              |
| Wing span       | 130.9 ft            |
| Wing area       | 685 ft <sup>2</sup> |

**Table 2 Airfoil configurations considered throughout the investigation**

| Airfoil   | Design point       | Boundary-layer constraint | Target $C_l$ | $M_\infty$ | $Re (\times 10^6)$ |
|-----------|--------------------|---------------------------|--------------|------------|--------------------|
| CA5488-85 | $\alpha = 0^\circ$ | $\theta_{\text{TE}}$      | 0.71         | 0.54       | 1.88               |
| CA5488-76 | $\alpha = 0^\circ$ | $\delta_{\text{TE}}^*$    | 0.71         | 0.54       | 1.88               |
| CA5427-72 | $L/D_{\max}$       | $\theta_{\text{TE}}$      | 1.12         | 0.54       | 1.27               |
| CA5427-76 | $L/D_{\max}$       | $x_0$                     | 1.12         | 0.54       | 1.27               |
| CA5410-85 | $\alpha = 0^\circ$ | $\theta_{\text{TE}}$      | 0.75         | 0.54       | 10.00              |
| CA1000-89 | $\alpha = 0^\circ$ | $\theta_{\text{TE}}$      | 0.58         | 0.10       | 1.00               |

freestream Mach numbers if the pressure rise induced by transonic shocks is effectively integrated with the APG produced by the subsequent recovery [29,38].

Similarly, the CA5427 airfoils were designed based on the operating conditions of the RQ-4B Global Hawk while considering a lower Reynolds number that could be produced during experimental testing. The target  $C_l$  was based on the  $L/D_{\max}$  for the LRN1015, as this parameter would coincide with the best loiter condition. Additionally, the lower surface was configured with a forced transition at  $0.70c$  and also remained fixed throughout the optimization. This design choice was implemented to maintain a similar lower surface distribution among the other design cases considered in this study which intrinsically helped to achieve the high  $C_l$  constraint without requiring significant modification of the lower surface. To showcase the effect of  $\theta_{TE}$  on the profile drag minimization, the CA5427-72 was optimized with minimum  $\theta_{TE}$ , while the CA5427-76 was purposefully designed with a Stratford pressure recovery to assess the airfoil design using the asymptotic limits of attached flow across the recovery region.

While not discussed in this paper, two additional airfoils were developed to illustrate the applicability of the method across a broader range of freestream operating conditions. Therefore, the parameters for the CA5410-85 and the CA1000-89 were simply selected to represent an expanded envelope of high  $Re$  and incompressible  $M$  conditions, respectively. Target  $C_l$  values were also generated using MSES at the corresponding operating conditions of interest. Both airfoils were optimized for minimum  $\theta_{TE}$  and with forced transition across the lower surfaces at  $0.50c$ , which were,

respectively, held fixed throughout the airfoil optimization process. The authors of Refs. [38,40] discuss in detail the specific design considerations, operating implications, and performance predictions for these two airfoils.

### B. CA5427-72: Airfoil Generation

A design point that targeted the  $L/D_{\max}$  condition of the baseline LRN1015 airfoil was of interest in this study since it represented the best loiter condition of the RQ-4B. In addition, a lower chordwise  $Re$  than that expected for actual flight was used in the development of this airfoil to better match experimental conditions that could be achieved in a wind tunnel environment. Following the process outlined in Sec. II.C, a series of  $C_p$  distributions were generated with varying  $x_0$  locations from  $0.60c$  to  $0.76c$  at intervals of  $0.01c$ . The resulting  $C_p$  distribution sweep is presented in Fig. 5a, along with the corresponding boundary-layer predictions in Figs. 5b–5d.

The minimum  $\theta_{TE}$  condition in Fig. 5b was identified for the  $x_0 = 0.72c$  distribution and was used to produce the CA5427-72 airfoil. Figure 6 shows the resulting geometry of the CA5427-72 alongside its boundary-layer characteristics, also compared to data for the LRN1015 at the same flow and angle-of-attack conditions. Variations between the design target and MSES data, particularly across the  $\delta^*$  and  $C_{f,x}$  distributions, are mostly attributed to the differences in the numerical solution algorithms of both integral boundary-layer solvers. However, the other design cases considered displayed better agreement since the boundary layer was subject to lower loadings and less-aggressive recoveries. The airfoil  $n_{crit}$  value at the desired

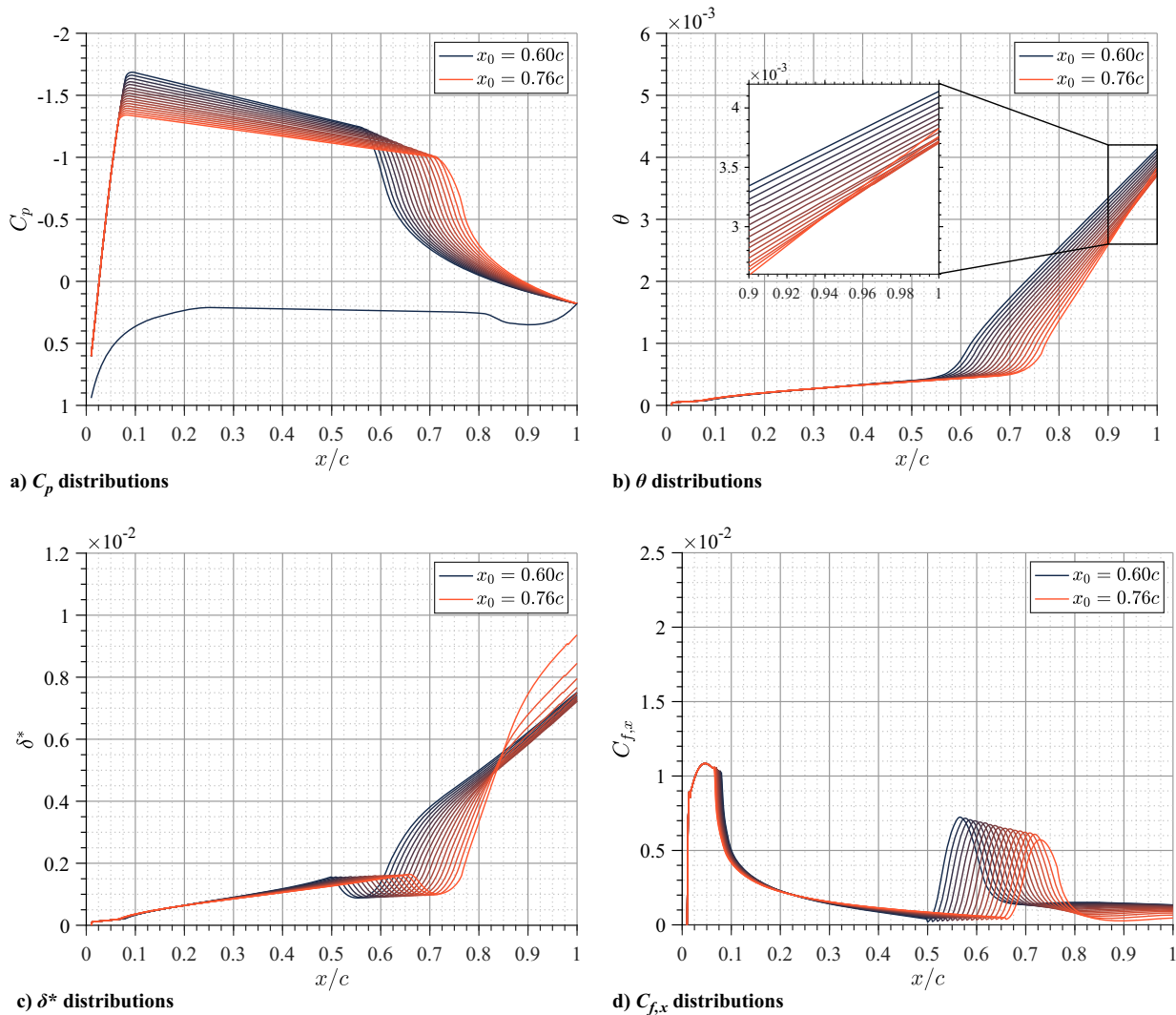


Fig. 5 Variations considered for the  $L/D_{\max}$  design point.

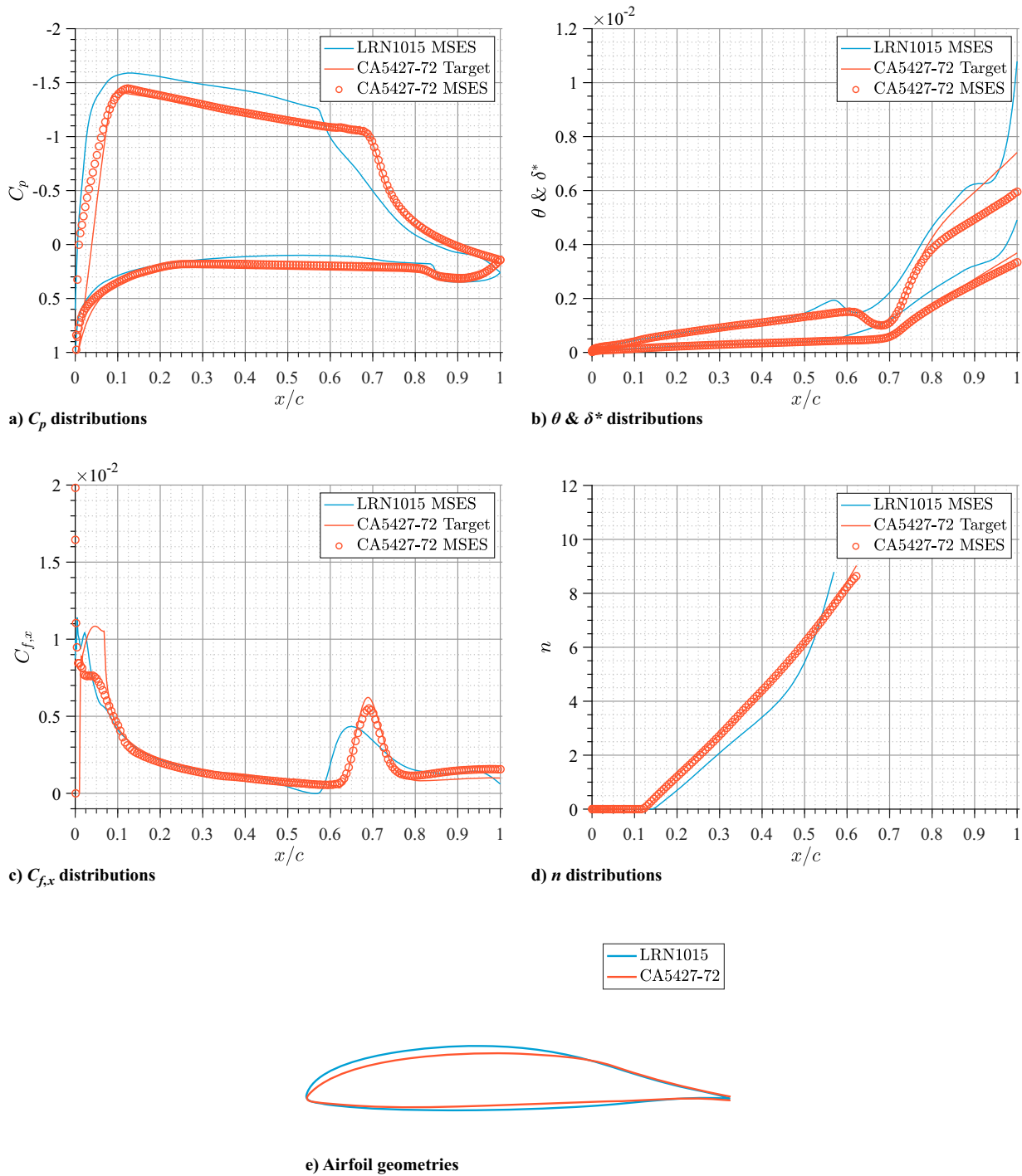


Fig. 6 CA5427-72 airfoil characteristics for minimum  $\theta_{TE}$  at the  $L/D_{max}$  design point.

transition location shows very good agreement with that predicted by the boundary-layer solver, as observed in Fig. 6d. The pressure distribution of the CA5427-72 airfoil in Fig. 6a produced a  $C_l$  value of 1.1256, showing excellent agreement with the target  $C_l$  condition of 1.12. It can also be observed in Fig. 6a that the CA5427-72 target and MSES  $C_p$  distributions show excellent agreement with minor variations near the leading-edge region. As was previously discussed in Sec. II.D, these observed differences are due to the smoothing required to ensure a continuous curve, especially near the leading edge of the airfoil.

At this design condition, the LRN1015 still displays an LSB, as can be identified in Fig. 6c by a small chordwise region displaying values of  $C_{f,x}$  below zero. The optimized CA5427-72 airfoil eliminates this separated region and also displays better trailing edge boundary-layer performance when compared to the LRN1015, as

observed in Fig. 6b. An interesting observation was made from Fig. 6a in which, without any prior knowledge, the upper-surface pressure distribution shape of the LRN1015 at the  $L/D_{max}$  condition closely resembled that of the optimized CA5427-72. This observed similarity provides confidence in the assumptions made for the design of the upper-surface pressure distribution outlined in Sec. II.C. The observation also suggests that, even though unconventional, the intentional use of an APG in certain flow conditions can lead to airfoils that produce low drag and feature extensive regions of laminar flow.

#### IV. MSES Performance Predictions

The following section presents the MSES performance predictions obtained for all the HALE-configuration airfoils designed using the present method and previously described in Sec. III.A. Special



emphasis is given to the airfoil operational implications and performance influences associated with the design conditions and target boundary-layer constraint.

#### A. The $\alpha = 0^\circ$ Design Condition

Figure 7 shows the lift, moment, and aerodynamic polar distributions produced for the CA5488-85 and LRN1015 airfoils using MSES. Considering the design point of  $\alpha = 0^\circ$ ,  $M = 0.54$ , and  $Re = 1.88 \times 10^6$ , with a constraint of  $C_l = 0.71$ , the CA5488-85 is observed to produce a  $C_d$  reduction of 9.06% and an  $L/D$  increase of 10.07% relative to the LRN1015 airfoil. This improved aerodynamic performance helps to validate the proposed methodology of associating the minimum drag condition with the  $\theta$  development across the surface of the airfoil. It can be determined that, at higher angles of attack immediately following the design condition, the airfoil contains regions of separated flow. This is reflected in the slope decrease along the  $C_l$  versus  $\alpha$  curve in Fig. 7a, in the sharp  $C_d$  increase observed in Fig. 7b, and was also confirmed in the off-design pressure distributions not shown. At these high angle-of-attack values, no appreciable difference is observed between the free- and fixed-transition analyses, especially when considering the airfoil  $C_d$ , which is more sensitive than other performance coefficients to transitional flow. This observation results from the  $C_p$  distributions displaying greater suction peaks and greater APGs as the angle of attack is increased. Hence, transition naturally occurs upstream of the design location that would be used to force transition in the solution. On the contrary, the lower angles of attack show significant

changes in the drag performance when comparing the fixed- and free-transition data. As the angle of attack is reduced, the severity of the APG across the upper surface is also reduced, and hence, if the transition is not fixed, the boundary layer is likely to separate and form an LSB at the onset of the pressure recovery. Nevertheless, even for the free-transition cases, the CA5488-85 airfoil shows slightly better performance for the lower region of the LRN1015 drag bucket. It should be noted that the off-design characteristics just described are associated with, and show the limitations of, point-design airfoils.

The MSES aerodynamic performance results for the CA5488-76 are shown in Fig. 8a, also compared to the LRN1015. The design point of  $\alpha = 0^\circ$ ,  $M = 0.54$ ,  $Re = 1.88 \times 10^6$ , and  $C_l = 0.71$  for this airfoil was observed to produce a  $C_d$  reduction of 6.19% and an  $L/D$  increase of 3.80%. The minimum  $\delta_{TE}^*$  condition represented by this airfoil shows an aerodynamic performance improvement, but the drag reduction is less than that observed for the CA5488-85, representing the minimum  $\theta_{TE}$ .

Aerodynamic polars for the fixed transition case of both CA5488-85 and CA5488-76 airfoils are presented in Fig. 9 to allow for direct comparison. The CA5488-85 is shown to produce marginally lower drag values when compared to the CA5488-76 across the drag bucket produced by these two airfoils. A smoother separation behavior towards a stalled condition can be inferred for the CA5488-76 as it shows a more gradual increase in drag across the upper limit of the drag bucket in Fig. 9. Similarly, the CA5488-76 achieves higher  $C_l$  values immediately following the design angle of attack when considering Figs. 7a and 8a. This behavior is expected as the CA5488-85

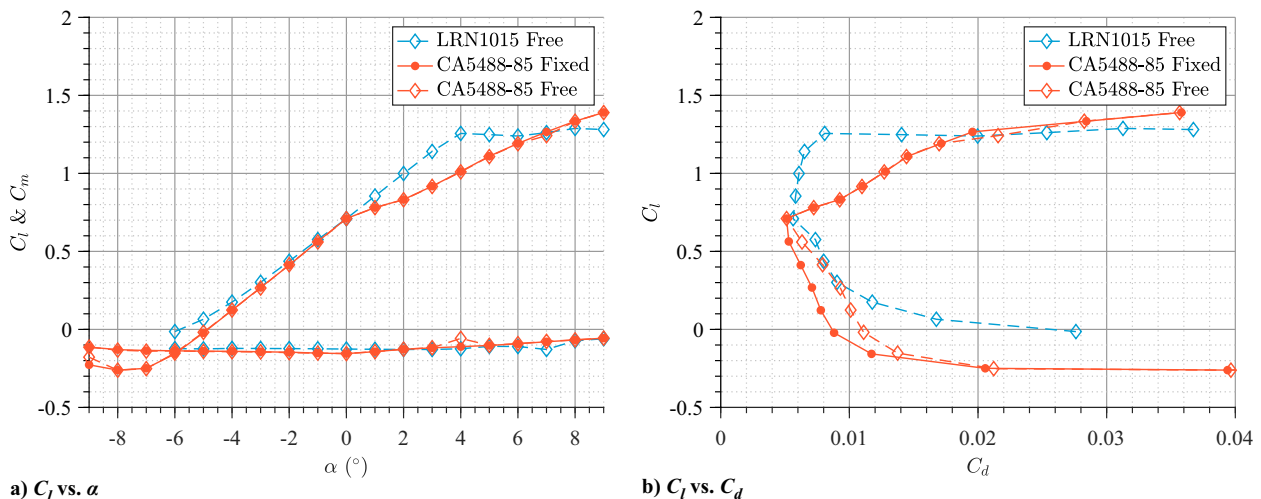


Fig. 7 Lift, moment, and aerodynamic polar curves for the CA5488-85 airfoil obtained using MSES.

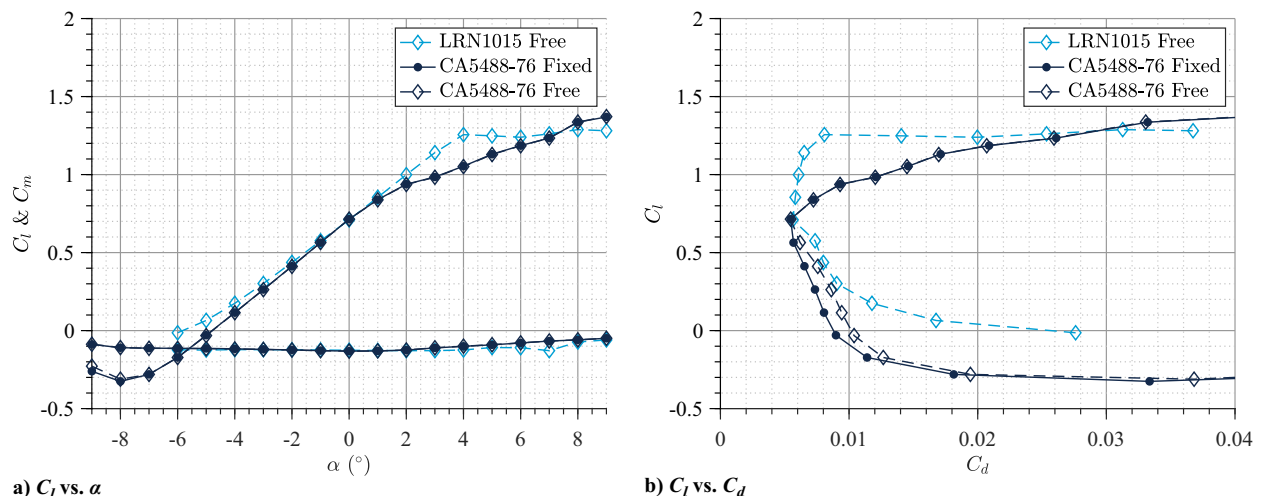


Fig. 8 Lift, moment, and aerodynamic polar curves for the CA5488-76 airfoil obtained using MSES.

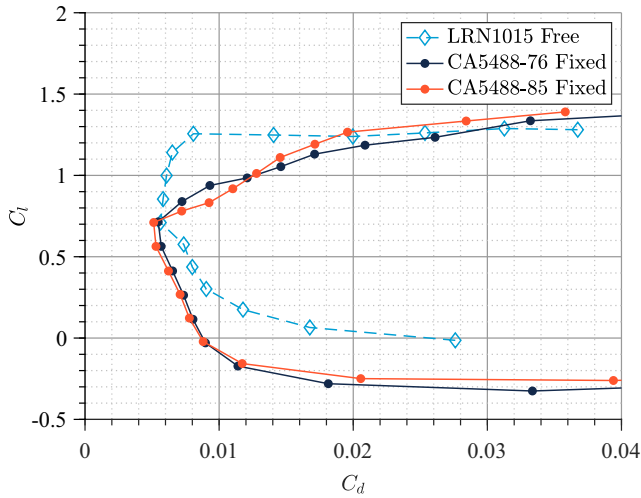


Fig. 9 Polar comparison for the CA5488 airfoil variants.

airfoil has a more aggressive recovery with a  $C_{f,x}$  distribution closer to zero. Hence, as compared to the CA5488-76, the boundary layer of the CA5488-85 has a limited ability to withstand further inertial opposition imposed by a more substantial off-design loading, resulting in earlier separation. Another interesting observation can be made when comparing the optimized polars to those of the LRN1015, which, as previously described, is associated with point-design limitations. Even though the design was performed at the same angle of attack for which the seed airfoil produced the target  $C_l$ , the design point results in the upper limit of the drag bucket for the optimized airfoils. Hence, even though the  $L/D$  for this angle of attack is improved when compared to the seed airfoil, the optimized airfoils could not achieve the  $L/D_{\max}$  produced by the LRN1015. Lastly, as can be observed in Figs. 7a and 8a, the pitching moment distributions for both optimized airfoils retain the same overall characteristics as those of the LRN1015.

### B. The $L/D_{\max}$ Design Condition

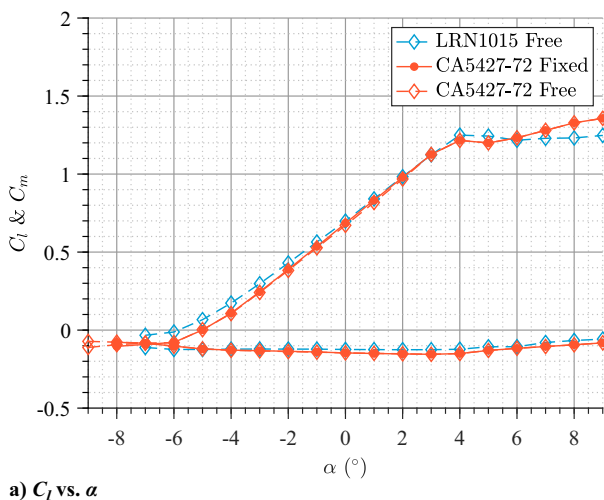
Aerodynamic data produced using MSES for the CA5427-72 and LRN1015 airfoils are presented in Fig. 10. The design  $\alpha$  for this airfoil was considered to be  $3^\circ$ , since it corresponded to the  $L/D_{\max}$  of the seed airfoil. At this angle of attack, the optimized CA5427-72 for the  $M = 0.54$ ,  $Re = 1.27 \times 10^6$ , and  $C_l = 1.12$  condition was found to produce a  $C_d$  reduction of 6.00% and an  $L/D$  increase of 6.59% when compared to the LRN1015. Similar to the observation made for the CA5488 airfoil variants in Sec. IV.A, the design point of the optimized airfoil also resulted in the  $L/D_{\max}$  condition and the

upper  $C_l$  edge of the drag bucket. Hence, for this particular airfoil, the optimized polar retains the same shape as that of the LRN1015, with the  $L/D_{\max}$  occurring at the same angle of attack. In addition, pitching moment characteristics were also observed to remain relatively unchanged when compared to the performance of the seed airfoil.

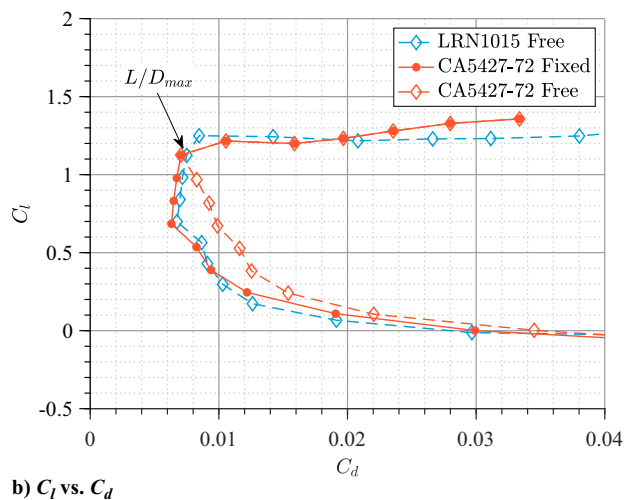
The airfoil is inferred to contain separated regions at conditions immediately above the design angle of attack, as was also discussed in Sec. IV.A, resulting from the imminent separation characteristic of Stratford-type recoveries. Another interesting observation resulting from this behavior is that, even though the curves for the optimized airfoil in Fig. 10 follow closely the trends produced by the LRN1015, the CA5427-72 fails to achieve the same  $C_{l_{\max}}$  as the LRN1015, as observed in Fig. 10a. Based on the previous observations made in Sec. IV.A, that would not likely be the case if the airfoil were optimized for a minimum  $\delta_{TE}^*$  condition which displayed better robustness to separation at off-design points. The polar region corresponding to angles of attack lower than the design point shows notable drag increases for the free-transition case, as was similarly identified in Sec. IV.A for the  $\alpha = 0^\circ$  design case. Nevertheless, the fixed-transition condition of the CA5427-72 in Fig. 10b displays notable drag reductions when compared to the LRN1015 across the  $C_l$  region of 0.39–1.13.

Figure 11 presents the MSES aerodynamic performance data obtained for the CA5427-76 airfoil, which are also compared to the data for the minimum  $\theta_{TE}$  optimized CA5427-72 airfoil. The CA5427-76 airfoil was purposefully designed to feature a Stratford recovery, as indicated in Table 2. Considering the design point in the aerodynamic polar of Fig. 11b, it can be observed that both airfoils achieve a very similar target  $C_l$  value, while the CA5427-76 produces a drag increase of 0.5 counts when compared to the CA5427-72 at  $C_l = 1.12$ . Even though it might be considered a negligible increment, it does provide validation to the proposed theory that, after a certain chordwise location, further laminar flow will offset the theoretical minimum profile drag that can be achieved.

The new airfoil with a shorter chordwise pressure recovery displays a stall at a lower angle of attack, as indicated by the sudden loss in  $C_l$  observed in Fig. 11a at  $\alpha = 4^\circ$ . This characteristic is attributed to the CA5427-76 airfoil having a pressure recovery closer to the theoretical Stratford distribution, which, as outlined by Liebeck [41,42], results in immediate separation when the airfoil is subjected to additional off-design loading. The  $C_d$  values for the CA5427-76 immediately following the design angle of attack  $\alpha = 3^\circ$  display greater drag values than the CA5427-72 in Fig. 11b due to the more aggressive separation characteristic just described. The lower angles of attack, however, display an interesting behavior for the fixed-transition cases in which the CA5427-76 achieves lower drag values than the CA5427-72, also on the order of less than 1 count, and a broadening of the drag bucket. As the airfoil loading across the upper



a)  $C_l$  vs.  $\alpha$



b)  $C_l$  vs.  $C_d$

Fig. 10 Lift, moment, and aerodynamic polar curves for the CA5427-72 airfoil obtained using MSES.

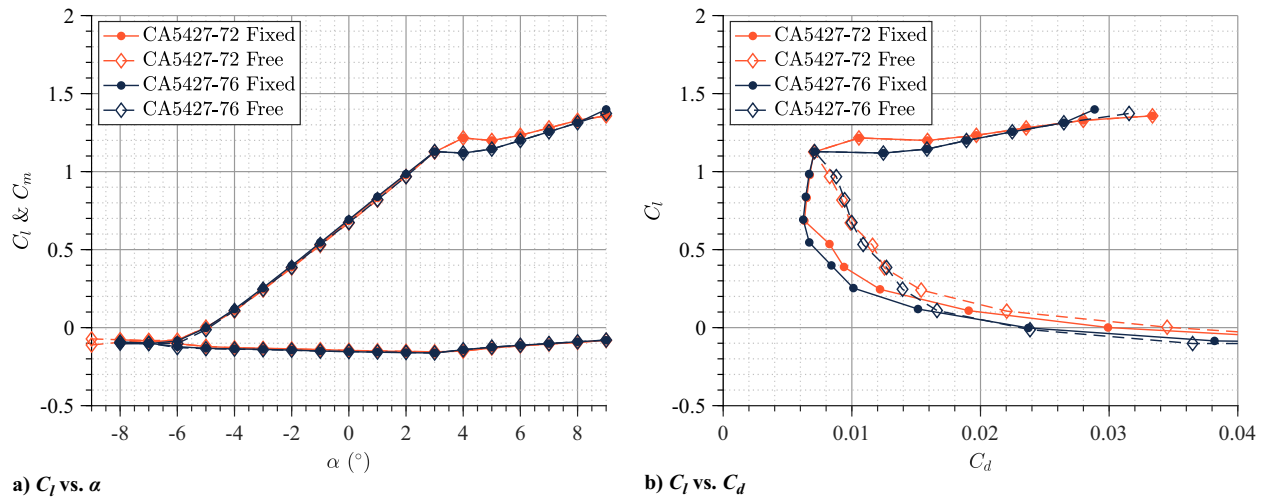


Fig. 11 Comparison of lift, moment, and aerodynamic polar curves for the CA5427-72 and CA5427-76 airfoils obtained using MSSES.

surface is reduced at these lower angles of attack, the additional laminar flow and less severe recovery result in a reduction in skin friction drag, with a relieved penalty of pressure drag associated with the aggressive recovery profile at the design point. This observation provides insights into the relationship between the optimal  $x_0$  and loading of the airfoil, as well as weighing different operational conditions when considering multipoint applications. Finally, the free transition data for the lower angles of attack (below the design  $C_l$  of 1.12) of the CA5427-76 airfoil display similar behavior to the CA5427-72 data, where substantial drag rises are observed.

## V. Experimental Validation of the CA5427-72 Airfoil

As previously discussed in Sec. III.A, the CA5427-72 airfoil was intentionally established for experimental validation of the design method being proposed. The following section describes the experimental methods and results obtained from this effort.

### A. Experimental Methods

#### 1. Experimental Airfoil Models

The CA5427-72 and LRN1015 airfoil geometries were used to design and build experimental models with a chord of 4.5 in., representative of the target Reynolds number conditions previously outlined in Table 2. Since airfoils generally require a finite-thickness trailing edge for manufacturing, the LRN1015 was modified to have a trailing edge thickness equal to 1% of the chord, blended upstream to the  $x/c = 0.50c$  location using the GDES function in XFOIL. No modification was required for the CA5427-72 airfoil since a finite trailing edge thickness of  $0.01c$  was already considered in its design. A multipiece assembly was considered for the design, which consisted of a main airfoil body that interfaced with supporting inserts on its sides. The main body, featuring a span of 5.2 in., was manufactured using SLA 3D printing, which allowed to internally route taps across the airfoil for surface pressure data acquisition. The pressure taps were configured with a 0.041 in. diameter and were arranged to interface with hypodermic tubing at both ends of the model. Additionally, the pressure taps were strategically swept along the span by considering a constant spacing to prevent premature boundary-layer transition from disturbances produced by upstream pressure taps. The placement of the pressure taps in this sweep configuration was also constrained to 50% of the semispan of the assembled model measured from the center to help mitigate sidewall influences that could affect the pressure measurements. In total, both models featured 47 pressure taps, with 23 and 22 distributed across the upper and lower surfaces, respectively, and two additional taps placed at the leading and trailing edges of the airfoil. The chordwise placement of the taps was purposely selected to capture the important pressure distribution characteristics of each airfoil surface under the design operating conditions.

Side inserts of the airfoil contours with a thickness of 0.25 in. were manufactured from stainless steel using wire electrical discharge machining (EDM). These inserts interfaced with the main body through threaded heat-set inserts that were previously installed. Additionally, a stainless steel shaft with a diameter of 0.375 in. was welded at the quarter chord location of each side insert, serving as a spar to control the angle of attack of the models during testing. The hypodermic tubing on the sides of the main body was also routed out of the wind tunnel through the inside of these shafts. Lastly, Teflon covers with 0.15 in. thickness were cut to match the airfoil profiles and glued on the external sides of the metal inserts to prevent damaging the wind tunnel windows. When fully assembled, each experimental model featured a span of 6 in. covering the entire length of the test section to help mitigate three-dimensional wing effects during testing. The details just described can be observed in Fig. 12 for the CA5427-72 experimental model.

As has been discussed in Secs. III.A and IV.B, the CA5427-72 airfoil was designed with forced transition across the lower surface and off-design performance implications with upper-surface forced transition were also considered. To experimentally achieve this forced transition condition, an array of vinyl trip dots with a diameter of 0.1 in. and center spacing of 0.2 in. was placed along the span of the experimental model at the desired chordwise location. The vinyl thickness of approximately 0.004 in. ensured that the trip dots produced the required disturbances to force bypass transition in the boundary layer.



Fig. 12 CA5427-72 experimental model.



## 2. Wind Tunnel Data Acquisition

Experiments were conducted at the Transonic Wind Tunnel Facility at the University of Illinois Urbana-Champaign. This suction-type, open-return wind tunnel features a test section with a cross-sectional area measuring 6 in.  $\times$  9 in. and running 18 in. in the streamwise direction. The test section is also fitted with porous walls having a 12% open-area ratio at a 60° inclination angle in a staggered configuration to mitigate interference effects and shock reflections [43]. Data acquisition was performed at the design Mach number of 0.54 and angle-of-attack range of  $-5^\circ$  to  $10^\circ$ . In total, three airfoil cases were tested, consisting of the LRN1015 and the CA5427-72 configured with free and forced transition.

A DTC Initium Data Acquisition system with an ESP-64HD Miniature Pressure Scanner with a 15 psid range was used for the acquisition of the airfoil surface and wake pressure measurements. A pitot-static wake traverse was installed in the test section to capture the pressure deficit in the wake generated by the airfoil models. The probe with an external diameter of 0.125 in. was located at the center of the test section and approximately 5 in. behind the airfoil model. Pressure measurements were acquired at 0.05 in. intervals at a frequency of 100 Hz and averaged across 6.35 s. The mean airfoil pressure measurements across all wake survey locations were used to obtain the lift and moment coefficients. Wake pressure data were then used to calculate the profile drag coefficient using the method of Betz in Ref. [44] after Pankhurst and Holder.

## 3. Uncertainty Analysis

As with all experimental investigations, the data acquired were associated with uncertainties that must be characterized to allow for proper interpretation of the results. Bias errors, associated with the uncertainty in measurement capabilities or accuracy of an instrument, were determined using the methods presented by Kline and McClintock [45] and Coleman and Steele [46]. Table 3 showcases sample uncertainties calculated for experiments conducted on the CA5427-72 airfoil and discussed in Sec. V.B, at its design operating conditions of  $M = 0.54$  and  $\alpha = 3^\circ$ . Most importantly, the uncertainties presented for all values lie within tolerable margins of the observations and comparisons made using the experimental data.

## B. Experimental Results for the CA5427-72 and LRN1015 Airfoils

### 1. Pressure Distribution at Design Angle of Attack

Figure 13 shows the pressure distributions obtained for the three airfoil configurations tested, consisting of the LRN1015, CA5427-72 transition-fixed, and CA5427-72 transition-free at the design conditions of  $\alpha = 3^\circ$  and  $M = 0.54$ . Table 4 also presents the calculated freestream and performance values corresponding to the experimental pressure distributions. While the observed Mach numbers for the three airfoils show excellent agreement with the design condition, the test Reynolds numbers consistently display lower values than the design target of  $Re = 1.27 \times 10^6$ . The target  $Re$  for this design case was calculated based on standard conditions in the laboratory environment. However, the experimental facility has no means of directly controlling the ambient temperature or pressure, which directly influences the freestream Reynolds number. As a result, the measured experimental Reynolds numbers shown were affected

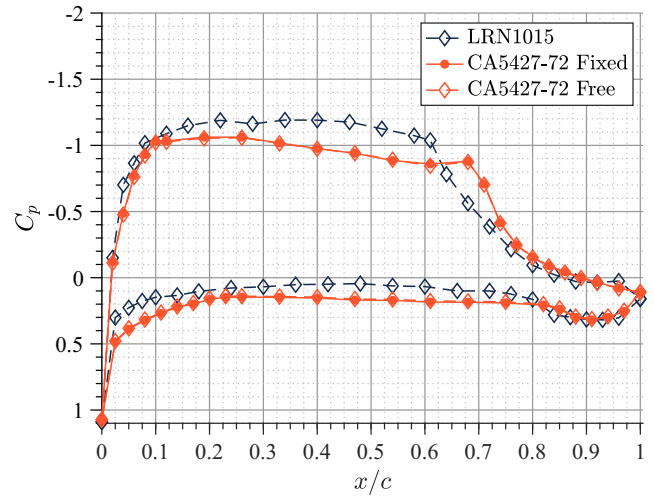


Fig. 13 Experimental  $C_p$  distributions for the CA5427-72 and LRN1015 airfoils at the design condition of  $M = 0.54$  and  $\alpha = 3^\circ$ .

Table 4 Experimental results for the CA5427-72 and LRN1015 airfoils at the design angle of attack of  $\alpha = 3^\circ$

| Airfoil         | $M$    | $Re(\times 10^6)$ | $C_l$  | $C_d$   | $C_m$   |
|-----------------|--------|-------------------|--------|---------|---------|
| LRN1015         | 0.5435 | 1.1772            | 0.8642 | 0.00874 | -0.1231 |
| CA5427-72 fixed | 0.5410 | 1.1687            | 0.8823 | 0.00738 | -0.1367 |
| CA5427-72 free  | 0.5467 | 1.1754            | 0.8713 | 0.00734 | -0.1342 |

by the atmospheric conditions and laboratory temperature at the time of testing. Nevertheless, the recorded  $Re$  values are within  $1 \times 10^5$  from the target, which still represent valid and acceptable flow conditions for which the intended design features of the airfoils can be demonstrated.

The experimental pressure distributions in Fig. 13 produce the general trends that were expected from the design predictions in Fig. 6a. For instance, the upper surface of both geometries follows the intended distribution contour with the CA5427-72 having more positive  $C_p$  values across the linear segment and a shorter recovery region, the lower surfaces achieve a very similar distribution, and a slightly lower trailing edge  $C_p$  value is observed for the CA5427-72 when compared to the baseline LRN1015. However, the experimental evaluation of both airfoils failed to capture the  $C_{l_{max}}$  conditions that were produced by MSES. This observed trait is a widely known limitation of potential-flow/integral boundary-layer coupled solvers and has been well documented by Coder and Maughmer [47,48]. In their study, they identified that all of the theoretical models considered, including MSES [48], consistently overpredicted the maximum lift coefficient of airfoils and that the differences observed were influenced by the airfoil geometry and operating flow conditions. Other studies in the literature that used MSES for similar airfoil analysis applications have also identified the same limiting characteristic [49–52]. As is later explained in greater detail, the wind tunnel environment was also associated with higher turbulent intensities than those considered in the design. The combined influence of these factors prevented the airfoil configurations from exactly producing the design distributions in Fig. 6a. Consequently, as shown in Table 4, the experimental  $L/D_{max}$  is reached at an operating condition of  $C_l = 0.87$ , as compared to the  $C_l = 1.12$  target.

With these considerations in mind, the analysis of the case study presented here was focused on the conditions for the experimentally achieved  $L/D_{max}$  parameter. This approach allowed us to directly assess the impact on  $L/D$  between the two airfoils and to identify performance benefits produced by the new airfoil design framework. Comparing analytical distributions at this experimental condition of interest can further assist in understanding the validity of the design environment towards the design objectives. Thus, the experimental

Table 3 Sample uncertainties for experimental measurements of the CA5427-72 airfoil at  $M = 0.54$ ,  $\alpha = 3^\circ$ , and fixed transition

| Parameter              | Reference value      | Absolute uncertainty        | Relative uncertainty, % |
|------------------------|----------------------|-----------------------------|-------------------------|
| $M$                    | 0.5410               | $\pm 1.1588 \times 10^{-3}$ | $\pm 0.2142$            |
| $Re$                   | $1.1687 \times 10^6$ | $\pm 2.6056 \times 10^2$    | $\pm 0.2230$            |
| $C_p$ ( $x/c = 0.54$ ) | -0.8853              | $\pm 4.2698 \times 10^{-3}$ | $\pm 0.4823$            |
| $C_l$                  | 0.8823               | $\pm 5.4236 \times 10^{-3}$ | $\pm 0.6147$            |
| $C_m$                  | -0.1367              | $\pm 2.5746 \times 10^{-3}$ | $\pm 1.8834$            |
| $C_d$                  | 0.0074               | $\pm 2.5844 \times 10^{-4}$ | $\pm 3.4925$            |
| $L/D$                  | 119.2297             | $\pm 4.2281$                | $\pm 3.5462$            |

$Re$ ,  $M$ , and  $C_l$  values recorded for each airfoil at their corresponding operating point of interest were used to generate MSES predictions, assuming the  $n_{crit} = 9$  design constraint. The resulting pressure distributions are presented in Fig. 14, all showing remarkable similarity to the experimental data. The MSES predictions indicate that the prescribed experimental  $C_l$  and flow operating constraints correspond to angles of attack of  $0.867^\circ$ ,  $1.346^\circ$ , and  $1.305^\circ$ , respectively, for the LRN1015, CA5427-72 fixed transition, and CA5427-72 free transition airfoil configurations. Similar results can be identified in the literature, where MSES often correlates  $C_l$  operating conditions to  $\alpha$  values that are lower than experimental observations [48,51]. These angle-of-attack differences appear to be case dependent, with increases in differences observed as the airfoil approaches stall.

When looking at the pressure distributions in Fig. 14 more closely, the predictions for the LRN1015 in Fig. 14a and CA5427-72 with fixed transition in Fig. 14b display recovery processes that begin at the intended design location and coincide with the experimentally observed recovery distributions. The drag predicted by MSES for the LRN1015 of  $C_d = 0.00777$  is 9.7 counts lower than the experimental value reported in Table 4. Similarly, the CA5427-72 fixed transition MSES prediction produced a drag coefficient of  $C_d = 0.00675$  which is just 6.3 counts lower than the experimental result. This disagreement between the experimental and predicted  $C_d$  values at the same  $C_l$  condition is not unexpected, as models based on potential-viscous coupled theory, such as MSES, are commonly known to underpredict drag [49–54]. Figure 14c, corresponding to the CA5427-72 free transition case, shows a predicted pressure recovery that begins downstream of the experimentally observed location and is preceded by a small region of constant  $C_p$ , suggesting

the presence of an LSB. A limited, separated region was confirmed at this location by a negative  $C_{f,x}$  observed in the simulated result, which produced a predicted  $C_d$  value of 0.00949. The disparity between the MSES predicted  $C_d$  and the experimentally calculated values in Table 4 for this particular case implies that the slight disagreement in  $C_p$  distributions between MSES and the experiment results in differences in the natural transition process at the condition being considered, or that transition is occurring at lower  $n_{crit}$  values corresponding to higher experimental turbulence intensities. If the latter is the case, the premature boundary-layer transition at the design condition might also be a contributor to the inability of the airfoils to achieve their maximum lift conditions observed in MSES. This prediction agrees with past observations made at the experimental facility, in which measured turbulence intensity values at various Mach numbers corresponded to  $n_{crit}$  values less than 9 [43]. Thus, a natural transition is achieved at the intended design location, but with a higher FPG than what was initially intended for the airfoil design.

To explore the impact of the critical amplification factor assumed for transition modeling, additional MSES analyses were performed using the CA5427-72 free transition experimental  $C_l$  data with varying  $n_{crit}$  values. Figure 15 shows the  $C_p$  distribution for the MSES prediction corresponding to  $n_{crit} = 3.05$ , which achieved an upper surface transition at  $x/c = 0.620$  matching the design constraint and shows better overall agreement with the experiment. The prediction driven by the experimentally achieved  $C_l$  corresponds to an angle of attack of  $1.235^\circ$  and produces a drag coefficient of  $C_d = 0.00700$ , just 2.5 counts higher than the fixed transition

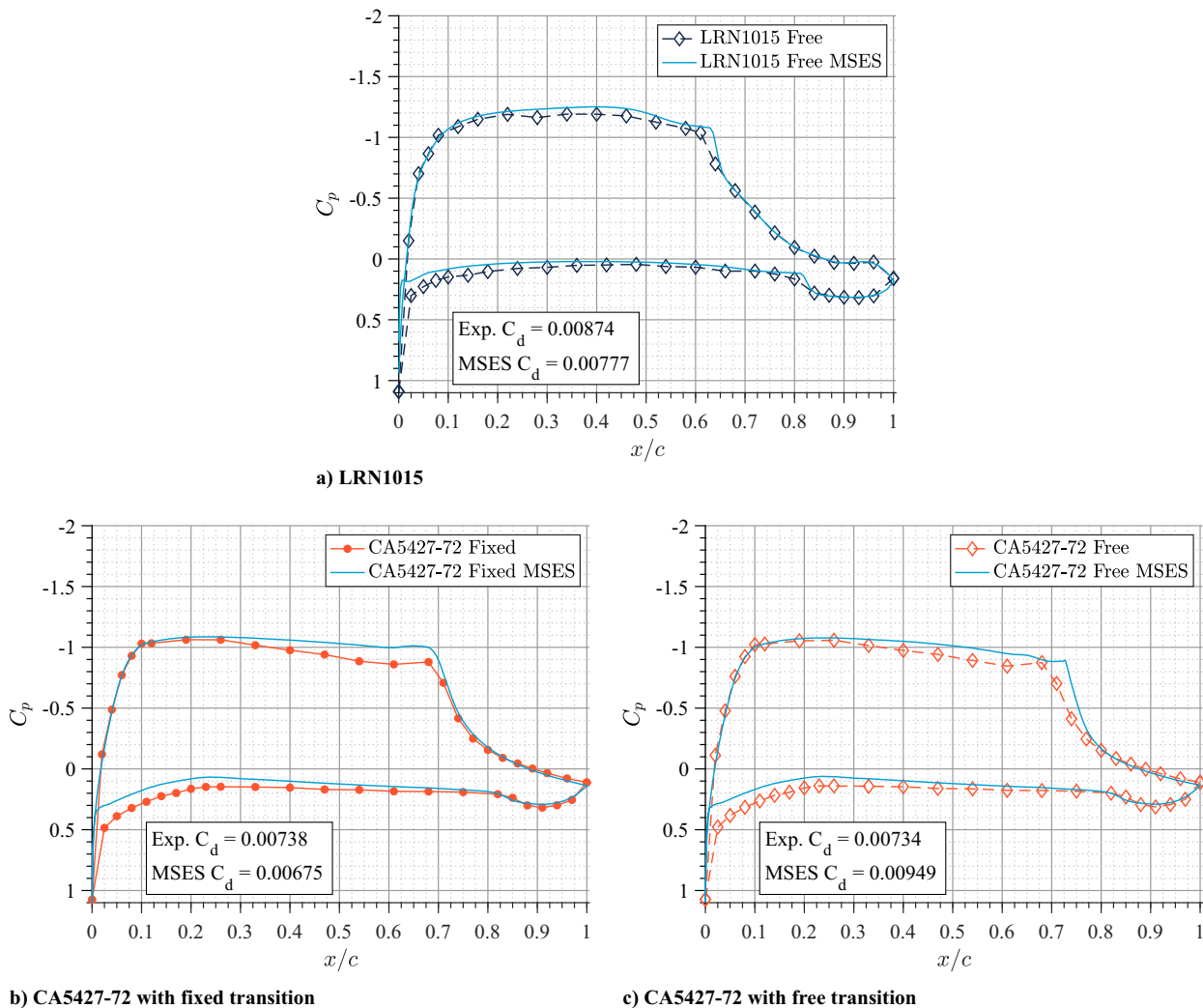
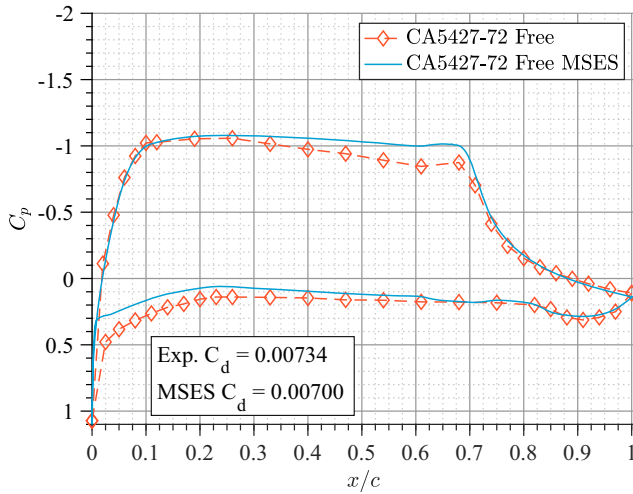


Fig. 14 MSES  $C_p$  predictions at the measured experimental conditions of each airfoil.



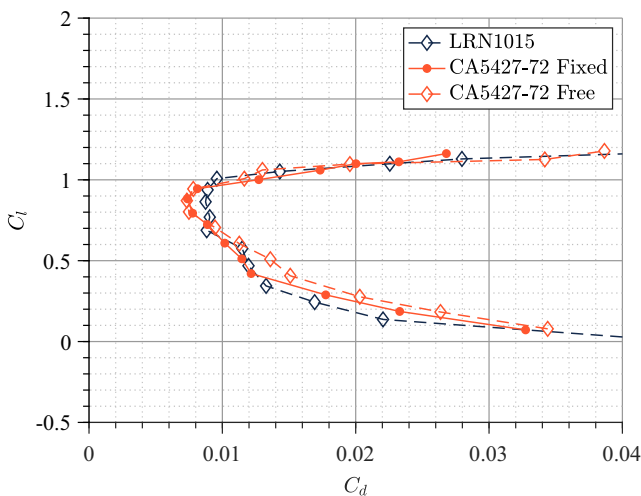


**Fig. 15** MSES  $C_p$  prediction for the CA5427-72 free transition case with  $n_{crit} = 3.05$ .

prediction in Fig. 14b. Similarities observed in the data produced by the pressure distributions in Figs. 14b and 15 imply that the turbulence intensity present in the wind tunnel environment does have a significant influence on the measured airfoil performance. The  $n_{crit}$  value of 3.05 corresponds to a turbulence intensity of 0.837%, which is higher but generally agrees with previously recorded values for the wind tunnel facility used [43]. This higher turbulence intensity is attributed to the motor operating speed required to produce the desired Mach number, which coincided with the switching boundary of the variable frequency drive and incurred in greater-than-normal velocity variations.

## 2. Aerodynamic Performance

Ventilated walls used in transonic wind tunnels generally have large interference influences on the effective angle of attack, to the extent that their experimental value is usually disregarded and the section normal force coefficient is emphasized [55]. Therefore, the experimental performance discussion for the LRN1015 and CA5427-72 airfoils is based on the aerodynamic polars presented in Fig. 16. Due to the inherent limitations associated with the porous test section, the  $C_l$  operating range achieved experimentally was also reduced relative to the MSES predictions in Fig. 10b. In particular, the low-drag bucket achieved for the optimized airfoil was limited to a  $C_l$  range between 0.7935 and 0.9450. Nevertheless, the experimental polar characteristics in Fig. 16 produced by the



**Fig. 16** Experimental aerodynamic polar curves for the CA5427-72 and LRN1015 airfoils.

three airfoils considered closely follow the outlined behaviors that were expected from the predictions in Sec. IV.B.

Both fixed- and free-transition configurations of the CA5427-72 displayed drag reductions of 15.56 and 16.01% over the LRN1015, respectively, at the intended design conditions. Similarly,  $L/D$  improvements of 20.91% for the fixed transition case and 20.05% for the free transition case were calculated. The measurable drag benefit obtained, which outperforms the MSES prediction, agrees with the same improvement trends that were outlined for this  $Re$  and  $M$  design condition during the airfoil development in Sec. IV.B. Additionally, the similarity in the data produced by the fixed- and free-transition configurations at this operating condition provides an indication of a natural boundary-layer transition occurring at or very near the design location.

Immediately above the design angle of attack, an abrupt drag rise is observed for the CA5427-72 airfoil when compared to the LRN1015. In agreement with the discussion in Sec. IV.B., this observation results from the imminent separation characteristics of Stratford-type recoveries previously described. Experimental data across this higher angle-of-attack region do not reveal a significant difference between the fixed- and free-transition cases. This characteristic is expected since the transition on the lower surface is fixed and more aggressive APGs across the upper surface result in upstream movement of the transition location from its design point. Therefore, the array of vinyl dots downstream of the transition location has no significant influence on an already turbulent boundary layer.

At the lower angles of attack, the fixed transition cases do display better drag performance across a wide  $C_l$  range, as expected from the predictions in Sec. IV.B. However, the free transition case does not immediately show a drag increase after the angle of attack is reduced from its design condition, as observed in Fig. 10b. This property is indicative of an inconsistency between the experimental observations and the ability of MSES predictions to effectively capture off-design transitional behavior and LSB development. Nevertheless, the  $C_d$  differences between the fixed and free transition cases that show improvement are in agreement with the MSES polar in Fig. 10b.

## VI. Conclusions

A new airfoil design methodology has been successfully introduced in which the boundary-layer physics were considered to optimize pressure distributions based on target constraints. Extension of the Squire–Young theory allowed the design framework to be used in identifying the pressure distribution corresponding to the minimum drag condition for the assumed pressure distribution. The proposed design theory has been verified through the development of several airfoil geometries considering HALE aircraft operating conditions as well as different target boundary-layer performance constraints. All airfoils showed better performance over the seed geometry at the design conditions, alluding to the advantages of a boundary-layer, performance-based design approach. The airfoils optimized to have minimum  $\theta_{TE}$  characteristics displayed the lowest drag when compared to the other pressure distributions generated through the same parameterization, suggesting a viable approach to the minimum drag solution for an airfoil based on boundary-layer theory. Other boundary-layer constraints revealed important implications when considering tradeoffs in drag minimization and off-design performance. For instance, the  $\delta_{TE}^*$  design example still produced a notable drag reduction over the seed geometry and displayed less aggressive stall characteristics than the  $\theta_{TE}$  case at the same operating conditions. This behavior resulted from the ability of a less aggressive recovery to accommodate stronger off-design pressure gradients. Based on these observations, it is also expected that a minimum shape factor constraint could be useful in achieving robust separation airfoils.

The experimental assessment of an optimized airfoil developed under this design framework indicated a 20.91% increase in  $L/D_{max}$  relative to the reference LRN1015 airfoil for a representative HALE aircraft application. This improved aerodynamic performance, as well as the observed pressure distribution design characteristics,

served to satisfactorily validate the design methodology. The performance differences between both geometries agreed with the MSES predictions but were found to be slightly higher, which was attributed to known characteristics of the flow solver. Fixed transition devices were observed to have an inconsequential influence on the design operating conditions. At off-design conditions, trends in the airfoil performance agreed with the MSES predictions made throughout the airfoil development process for both fixed- and free-transition cases. Discrepancies observed between the MSES predictions for the CA5427-72 fixed- and free-transition cases were attributed to differences in the natural transition location produced due to the likely presence of higher turbulent intensities in the testing environment.

While several optimized airfoils have been developed, the fundamental contribution of this study is intended to be the design ideology itself, in which a pressure distribution is intentionally configured to achieve desired boundary-layer characteristics that ultimately govern the drag produced by an airfoil. Through this approach, the boundary-layer integral parameters drive the inverse design process in lieu of the pressure distribution, which is iteratively modified until a desired performance is obtained. Furthermore, the methodology that has been introduced provides an increased physical understanding regarding the main performance drivers in an airfoil inverse design process that would otherwise be explicitly unknown in more conventional methods. It is also recognized that airfoil design involves the consideration of multiple operating conditions. Thus, the proposed framework should also be expanded to account for multipoint design applications where cost functions involving weighted integral boundary-layer parameter data across a range of conditions are considered. Additionally, the methodology should be assessed with common geometrical constraints (i.e., thickness and leading-edge radius) applied to airfoil designs that could limit the performance gains that were discussed. A more elaborate parametric variation of the pressure distribution should also be considered, in which functions can be introduced across different segments. This would allow to find optimum solutions independently across different regions and address off-design limitations through the intentional design of features such as transition ramps. Lastly, further consideration should be given to the natural transition location and its influence on optimal solutions and off-design performance.

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